

Hybrid Sparse-Dense Retrieval for EU Climate Law: Evaluating BM25, BGE-M3 and Reciprocal Rank Fusion on a Statute-Level Benchmark

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Abstract

Grounded responses in legal AI applications depend critically on first-stage retrieval: a system that fails to surface the correct statutory provision cannot produce a verifiable, citation-supported answer. We evaluate three retrieval configurations—BM25 lexical search, BGE-M3 dense retrieval (BAAI/bge-m3, 1,024dim), and a hybrid combining both via Reciprocal Rank Fusion (RRF, $k=60$)—on a manually constructed benchmark of 71 queries over EU Climate Law. The corpus comprises 1,156 article-level provisions from 72 EU instruments extracted from the EU CELLAR database via SPARQL and Formex XML. We measure retrieval effectiveness using HR@1, HR@5, HR@10, and MRR@ k with Wilson 95 % confidence intervals, and complement these with reliability (per-query hit patterns) and efficiency (index size and query latency) analyses. To the best of our knowledge, this is the first systematic comparative study of hybrid sparse-dense retrieval on EU statutory text, and the first statute-level retrieval benchmark for EU Climate Law. Experimental results are reported as [PLACEHOLDER] pending completion of the evaluation runs. The findings will directly inform the design of first-stage retrieval components in grounded legal AI systems where source attribution to specific CELEX instruments and articles is mandatory.

Keywords: EU Climate Law; legal information retrieval; hybrid retrieval; BM25; BGE-M3; reciprocal rank fusion; retrieval-augmented generation; statute-level benchmark.

1 Introduction

As a motivating hypothetical, consider a legal analyst querying a BM25-only system with “*What monitoring obligations apply to aircraft operators under the EU Emissions Trading System?*” A BM25-only system might return top-ranked provisions from Regulation (EU) 2015/757 on maritime emissions monitoring (CELEX 32015R0757) because the terms *monitoring*, *operators*, and *obligations* achieve high BM25 scores against that instrument’s operative articles. The correct provisions—Articles 14–15 of Directive 2003/87/EC (32003L0087)—could rank below position 10 owing to aviation-specific vocabulary (*tonne-kilometre data*, *verified emissions report*) that is absent from the query. The downstream language model, receiving context that contains no mention of aviation or Article 14, would be unable to produce a grounded answer and would either hallucinate a response or correctly invoke an insufficient-context guard.

This hypothetical scenario illustrates why first-stage retrieval quality is the rate-limiting factor for grounded legal AI applications. Retrieval-augmented generation (RAG) pipelines for statute interpretation require that the retriever surface not merely topically similar text, but the legally binding provisions that *answer* the query. In EU Climate Law, this requirement is compounded by legislative density (72 enacted instruments), cross-instrument definitional dependencies (e.g.,

the CBAM Regulation cross-references ETS allowance concepts defined in Directive 2003/87/EC), highly technical domain-specific vocabulary (*free allocation, carbon leakage, corrigendum*), and the urgent policy context that makes error-free source attribution a professional and legal necessity.

Sparse lexical retrieval (BM25) captures exact statutory terminology but is susceptible to vocabulary mismatch [1]. Dense semantic retrieval (BGE-M3) generalises beyond exact matches but may conflate legally distinct concepts sharing surface similarity [2]. Hybrid fusion via Reciprocal Rank Fusion (RRF) combines both lists without requiring score normalisation or labelled training data [3], making it particularly suitable for a domain where annotated retrieval pairs are scarce.

Research Questions.

RQ1. Does hybrid sparse-dense retrieval (BM25 + BGE-M3 with RRF $k=60$) improve the effectiveness and reliability of first-stage grounding for EU Climate Law compared with BM25-only and BGE-M3-only retrieval?

RQ2. Is first-stage hybrid retrieval sufficient to support accurate and domain-relevant grounded responses in EU Climate Law, given retrieval coverage, hard-query failure rates, and efficiency overhead?

Contributions. This paper makes four contributions:

1. A 71-query manually verified benchmark for EU Climate Law statute-level retrieval with binary CELEX-level relevance judgements (§4.6).
2. A reproducible article-level corpus of 1,156 provisions from 72 EU climate instruments, extracted from the EU CELLAR SPARQL endpoint via a documented two-phase pipeline (§4.1).
3. A three-way comparative evaluation of BM25, BGE-M3, and BM25+BGE-M3 RRF across effectiveness, reliability, and efficiency dimensions (§7).
4. A characterisation of anticipated first-stage retrieval failure modes and their implications for reliable source attribution in domain-specific LLM deployments (§9).

2 Related Work

2.1 Lexical Retrieval in Legal IR

BM25 remains the dominant first-stage lexical retrieval model owing to its probabilistic grounding and robust performance without training data [1]. In legal retrieval, exact match over statutory vocabulary is particularly valuable: defined terms such as *installation, allowance, and tonne-kilometre* carry precise legal meanings that synonymy-based generalisation may distort. Mori and Kano [4] examine statute-level retrieval for legal question answering and find that lexical signals remain strong for definitional and obligation-type queries where the relevant provision shares key vocabulary with the question. The NOWJ team at COLIEE [5] demonstrate that well-tuned BM25 baselines are competitive on statutory entailment tasks, establishing a high bar for learned alternatives. The fundamental limitation of pure lexical retrieval—term mismatch between user query vocabulary and statutory vocabulary—motivates the dense component evaluated in this paper.

2.2 Dense Retrieval and Legal Embeddings

Dense retrieval encodes both queries and documents into a shared embedding space, enabling semantic matching beyond exact term overlap [2]. The BGE-M3 model achieves leading performance on the MTEB multilingual retrieval benchmark through self-knowledge distillation and multi-granularity encoding [2]; critically for this work, its 8,192-token context window accommodates long legislative articles without arbitrary truncation for the majority of the corpus. Rayo et al. [6] evaluate dense retrieval for regulatory compliance queries and identify distributional shift between general-domain pre-training corpora and EU statutory text as a factor limiting performance. LegalRAG [7] integrates dense retrieval into a legal QA pipeline and reports that semantic matching reduces hard failures on queries whose vocabulary diverges from the document collection. A shared limitation is that dense models may retrieve thematically related but legally distinct instruments—a failure mode characterised in §9.

2.3 Sparse Learned Representations

SPLADE [8] learns sparse representations by applying a log-saturation activation over BERT vocabulary logits, producing inverted-index-compatible vectors that encode both lexical and expanded signals. The SPLADE-v2 extension [9] improves effectiveness through distillation from a dense teacher and hard-negative sampling, substantially narrowing the gap between sparse learned models and dense retrievers on standard BEIR benchmarks. However, both variants require domain-specific fine-tuning to be effective: without in-domain annotated pairs, SPLADE models exhibit vocabulary hallucination—expanding queries towards general-domain concepts absent from the target corpus. No publicly available SPLADE model trained on EU statutory text is known to the authors. This gap justifies using terminology-preserving BM25 as the lexical component, and flags SPLADE fine-tuning on EU law as a productive direction for future work.

2.4 Hybrid Retrieval and Rank Fusion

Hybrid retrieval combining sparse and dense ranked lists via Reciprocal Rank Fusion was shown by Cormack et al. [3] to outperform individual rank learning methods and Condorcet fusion on TREC collections without requiring score normalisation. HyPA-RAG [10] applies hybrid parameter-adaptive retrieval to AI legal and policy applications. LRAGE [11] evaluates multiple hybrid retrieval configurations in a legal RAG benchmark framework. A recurring theoretical motivation across these systems is that the independence assumption underlying RRF—that ranking errors of each retriever are partially uncorrelated—may hold in legal IR because lexical and semantic failure modes are empirically distinct; this is a testable claim that §6.2 evaluates.

2.5 Legal RAG Systems

Several systems address the end-to-end legal RAG pipeline. CBR-RAG [12] augments retrieval with case-based reasoning, using structural similarity between retrieved precedents and the current query as a re-ranking signal. LexRAG [13] combines lexical retrieval with generation for legal reasoning tasks and reports that lexical precision is a stronger predictor of answer correctness than semantic similarity on statutory interpretation queries. LexDrafter [14] applies RAG to legislative drafting assistance, finding that article-level chunking aligns with drafting practice better than fixed-token windows. Branting et al. [15] exploit legal document structure (recitals, chapters, articles, sub-paragraphs) as a retrieval signal, reporting gains on tasks where the correct provision is embedded in a specific structural position. A consistent gap across this body of work is the absence of systematic evaluation on EU statutory text; most systems focus on common-law jurisdictions (US, UK) or case law retrieval, with EU civil-law legislation—which has distinct structural and terminological properties—largely unexplored.

2.6 Retrieval Reliability and Source Attribution

Ravichander et al. [16] identify retrieval reliability—the consistency with which a system surfaces the correct provision across paraphrased variants of the same query—as a distinct evaluation dimension from single-query effectiveness, and argue that unreliable retrieval is a principal driver of hallucination in grounded legal AI. Zheng et al. [17] frame legal retrieval as a reasoning problem in which the first-stage retriever must resolve implicit definitional dependencies between instruments before a chain-of-thought generator can be applied. Both works underscore that for legally consequential applications—where every claim in an answer must be attributed to a specific, verifiable statutory provision—single-point IR metrics understate the practical risk introduced by retrievers with uneven per-query coverage. We address this gap directly through the reliability dimension in §6.2.

2.7 Evaluation Frameworks for Legal Retrieval

The COLIEE benchmark [5] is the most widely used legal retrieval evaluation, covering statute retrieval and legal entailment tasks primarily over the Japanese Civil Code and Canadian case law. LRAGE [11] proposes a framework for evaluating legal RAG systems that decomposes performance into retrieval coverage, attribution accuracy, and answer faithfulness. HyPA-RAG [10] includes an evaluation component for legal and policy question answering. None of these frameworks provides a benchmark for EU Climate Law statute retrieval, where the target collection spans 72 instruments in the CELLAR repository and queries must resolve cross-instrument definitional dependencies. The benchmark constructed in this paper fills this gap and is designed to be directly reproducible from the publicly available CELLAR SPARQL endpoint.

3 Research Gap and Contributions

The Related Work establishes three specific gaps that this paper addresses:

Corpus gap. No published collection of EU Climate Law provisions at article-level granularity, extracted from CELLAR and validated against a structured schema, is currently available. The corpus assembled here (1,156 provisions, 72 instruments) provides the first reproducible baseline for this domain.

Benchmark gap. No manually constructed query benchmark for EU Climate Law statute-level retrieval exists. Existing benchmarks (COLIEE, LRAGE) address different jurisdictions and document types. The 71-query benchmark developed here uses binary CELEX-level relevance judgements calibrated to EU legislative practice.

Evaluation gap. Although hybrid retrieval has been studied in general and legal IR, no prior work has systematically compared BM25, BGE-M3, and RRF specifically on EU statutory text under a shared evaluation protocol that reports Wilson confidence intervals, per-query reliability, and efficiency simultaneously.

These gaps jointly define a contribution that cannot be addressed by re-running any existing system on any existing benchmark.

4 Methodology

Figure 1 gives an overview of the full pipeline. The quantitative evaluation covers the shaded retrieval components; the generation stage is described for completeness but evaluated only qualitatively.

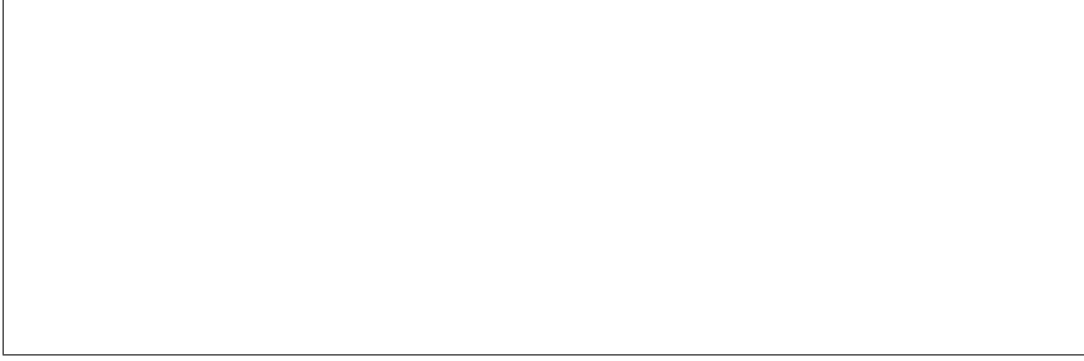


Figure 1: System architecture. A query is processed concurrently by BM25 and BGE-M3; the two ranked lists of 20 candidates each are merged by RRF to produce a fused top-5 context. The fused context is passed to Llama 3.3-70B (Ollama, local inference) for strictly-grounded answer generation, with every claim required to cite [CELEX_ID -- ArticleN]. Quantitative evaluation covers only the retrieval components (shaded region).

4.1 Corpus Construction

The corpus is extracted from the EU CELLAR repository using a documented two-phase SPARQL pipeline.

Phase 1 — Batch SPARQL. A SPARQL query against the CELLAR endpoint retrieves all work URIs associated with 21 EuroVoc climate and energy concepts, yielding 194 unique CELEX identifiers for Directives and Regulations.

Phase 2 — Per-document manifestation retrieval. For each work URI, a second SPARQL query resolves the available Formex XML (`fmx4`) manifestation URL; where no Formex manifestation exists, an HTML fallback is used. Documents are fetched with retry and exponential back-off and cached locally.

Parsing. Formex XML is parsed with `lxml`. Articles are identified by the `<ARTICLE>` element; article text is reconstructed from `<ALINEA>` leaf nodes to avoid duplication of numbered subparagraphs inherited from parent elements. Each article is serialised to a JSON record `{celex_id, doc_type, article_number, article_text, cross_references, concept_ids}` and validated against a Pydantic schema before inclusion; records containing placeholder text are discarded.

Corpus statistics. The final corpus contains **1,156** article-level provisions from **72** CELEX instruments (Directives and Regulations), covering ETS, CBAM, Taxonomy, LULUCF, F-Gas, Maritime MRV, ESR, the European Climate Law, Energy Efficiency, CSDDD, and the Governance Regulation. Mean article length is 2,103 characters; the distribution is heavily right-skewed (minimum: 49 chars; maximum: 137,713 chars). The maximum article exceeds BGE-M3’s 8,192-token context window; truncation behaviour for this extreme tail is acknowledged as a limitation in §10.3.

Why article-level chunking. EU legislative articles are drafted as semantically self-contained units: each article establishes one obligation, one definition, or one procedural rule. Article-level chunking avoids the arbitrary window-boundary decisions of token-based chunking and produces retrieval units that align with how lawyers reason about statute—specifically, the citation format [CELEX_ID -- ArticleN] that downstream grounded generation requires.

4.2 Sparse Retrieval — BM25

The sparse retriever implements BM25Okapi (rank-bm25 0.2.2) with $k_1=1.5$ and $b=0.75$. A custom legal tokeniser applies the regex pattern `[a-zA-Z0-9](?:[a-zA-Z0-9\-\_]*[a-zA-Z0-9])?` before lower-casing. This preserves hyphenated statutory terms (*net-zero*, *CBAM-certificate*, *cross-border*) as single tokens. No stemming and no stopword removal are applied: EU legal definitions are definitionally precise in ways that stemming distorts—*emission* (singular) and *emissions* (plural) are used with distinct intent in ETS legislation, and removing stopwords can break fixed legal phrases such as *free of charge* and *not later than*.

The index is built by fitting BM25Okapi over all 1,156 tokenised article texts and serialised via `pickle`. At query time, BM25 scores all documents and returns the top 20 ranked results.

4.3 Dense Retrieval — BGE-M3

The dense retriever uses BAAI/bge-m3 loaded via the `sentence-transformers` library [18, 2]. The model produces 1,024-dimensional vectors.

Model selection rationale. BGE-M3 was selected for three reasons: (1) it is fully open-weight and requires no API key, making results reproducible without cost; (2) its 8,192-token context window accommodates the vast majority of EU articles, which average 2,103 characters (≈ 525 tokens); (3) it achieves state-of-the-art performance on the MTEB multilingual retrieval benchmark, providing a strong semantic baseline without domain-specific fine-tuning.

Indexing. Documents are encoded with `encode_corpus(texts, normalize_embeddings=True, batch_size=64)`. Corpus vectors are L2-normalised and loaded into a FAISS `IndexFlatIP`; inner product on normalised vectors is equivalent to cosine similarity.

Retrieval. The query is encoded with `encode_queries([query], normalize_embeddings=True)`, producing a single normalised vector. FAISS returns the top 20 nearest neighbours by inner product as exact nearest-neighbour search.

4.4 Reciprocal Rank Fusion

Retrieval lists from BM25 and BGE-M3 are merged via Reciprocal Rank Fusion. For each document d , the fused score is:

$$\text{RRF}(d) = \sum_{r \in \{\mathcal{R}_s, \mathcal{R}_d\}} \frac{1}{k + \text{rank}_r(d)} \quad (1)$$

where $k=60$ is the smoothing constant [3] and $\text{rank}_r(d)$ is the 1-based rank of document d in retriever list r (∞ if d is absent from r). Documents are deduplicated by the composite key `celex_id::article_number` before fusion. The top 5 fused results ($k_f=5$) are returned as `FusedResult` objects carrying the RRF score, individual dense and sparse ranks, and the article text.

Why RRF over learned fusion. Two properties make RRF appropriate here. First, BM25 scores are unbounded positive reals while cosine similarities lie in $[-1, 1]$; combining raw scores requires normalisation assumptions that may be violated. RRF operates on ranks only and is therefore scale-invariant [3]. Second, no labelled relevance pairs for EU Climate Law retrieval exist in any public training set; learned fusion weights require supervised signal that is unavailable.

Concurrent execution. Both retrievers are invoked concurrently via `ThreadPoolExecutor(max_workers=2)`. If one retriever raises an exception, the controller logs a warning and continues with the remaining list; if both fail, the exception propagates.

Grounding hypothesis. Because BM25 and BGE-M3 are expected to fail on largely disjoint query subsets (vocabulary mismatch vs. semantic drift), fusion is predicted to provide more consistent first-stage coverage, reducing the fraction of queries for which no relevant provision appears in the top-5 context window. RQ1 tests this hypothesis empirically.

4.5 Algorithm

Algorithm 1 formalises the full retrieval pipeline.

Algorithm 1: Hybrid Sparse-Dense Retrieval Pipeline

Input: Query q ; per-retriever limit $k_r=20$; fusion output $k_f=5$; RRF constant $k=60$
Output: Fused list $\mathcal{R}_f$, $|\mathcal{R}_f|=k_f$

```
// Step 1 -- concurrent retrieval (ThreadPoolExecutor, max_workers=2)
 $\mathcal{R}_s, \mathcal{R}_d \leftarrow \text{parallel}\{\text{BM25.RETRIEVE}(q, k_r), \text{BGE-M3.RETRIEVE}(q, k_r)\}$ 

// Step 2 -- RRF fusion (Equation 1)
 $\mathcal{D} \leftarrow \text{DEDUPLICATE}(\mathcal{R}_s \cup \mathcal{R}_d)$  // key: celex_id::article_number
foreach  $d \in \mathcal{D}$  do
     $\text{rrf}(d) \leftarrow \sum_{r \in \{\mathcal{R}_s, \mathcal{R}_d\}} \frac{1}{k + \text{rank}_r(d)}$ 
end

// Step 3 -- return top- $k_f$  by RRF score
return  $\text{TOPK}(\mathcal{D}, k_f; \text{key} = \text{rrf})$ 
```

4.6 Evaluation Benchmark Construction

The benchmark comprises 71 manually constructed queries covering the eleven EU Climate Law instrument families in the corpus. Each query is phrased as a natural-language legal question of the type encountered in regulatory compliance, policy analysis, or statutory interpretation.

Relevance judgements. Relevance is binary at the CELEX instrument level (one or two relevant documents per query). A CELEX ID is judged relevant if the instrument contains the provision or provisions that directly and materially answer the query; thematically related instruments that do not contain the operative provision are judged not relevant.

Corrigendum normalisation. CELEX identifiers carrying a corrigendum suffix (e.g., 32003L0087R(02)) are matched against gold identifiers after stripping the `R(...)` suffix, so that corrigendum-form identifiers in the corpus correctly match base-form identifiers in the benchmark.

Statistical note. At $n=71$ queries the Wilson 95 % confidence interval width for HR@5 is approximately ± 0.06 – 0.12 depending on the observed rate (Equation 4), which is sufficient to distinguish systems separated by ≥ 0.15 in HR@5.

Why document-level retrieval. Legal citations refer to instruments and articles, not sub-article passages. Document-level (CELEX-level) matching is the appropriate proxy for grounded answerability: if the correct instrument is not retrieved, the correct article cannot appear in the LLM’s context.

5 Experimental Setup

Hardware. [Specify machine.]

Software.

- Python [version]
- `sentence-transformers` $\geq 3.3.0$ [18]
- `faiss-cpu` 1.14.2
- `rank-bm25` 0.2.2
- `numpy` 2.4.6
- `pydantic` 2.13.4
- Llama 3.3-70B-Instruct via Ollama [version] (qualitative demonstration only)

All library versions are pinned in `requirements.txt`. The BGE-M3 model weight hash is recorded at download time to ensure reproducibility.

Index build. The BM25 index is built by fitting `BM25Okapi` over all 1,156 tokenised article texts and serialised via `pickle`. The dense FAISS index is built by embedding all articles in batches of 64 with `encode_corpus`, L2-normalising the resulting $1,156 \times 1,024$ matrix, and writing to disk via `faiss.write_index`. Build times are reported in Table 2.

Query latency measurement. For each retrieval condition, all 71 gold queries are executed in five independent runs; median and 95th-percentile latency are reported. Times include tokenisation, score computation, and result assembly but exclude index loading from disk.

6 Evaluation Framework

6.1 Dimension 1: Effectiveness

Hit Rate at k . $\text{HR@}k$ is the fraction of queries for which at least one relevant CELEX identifier appears in the top- k results:

$$\text{HR@}k = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\text{rank}_q \leq k] \quad (2)$$

where rank_q is the rank of the highest-ranked relevant document for query q and $|Q|=71$. We report $k \in \{1, 5, 10\}$.

Mean Reciprocal Rank. $\text{MRR@}k$ is the mean of the reciprocal rank of the first relevant document:

$$\text{MRR@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q} \cdot \mathbf{1}[\text{rank}_q \leq k] \quad (3)$$

We report $k \in \{5, 10\}$.

Note on nDCG. The benchmark provides binary relevance judgements. nDCG with binary labels differs from MRR only in how sub-optimal ranks are weighted; it adds no analytical information beyond what $\text{HR@}k$ and $\text{MRR@}k$ already capture and is therefore omitted.

Confidence intervals. Wilson 95 % confidence intervals are reported for all HR@ k values:

$$CI_{95\%}(\hat{p}) = \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \quad (4)$$

where $\hat{p}$ is the observed hit rate, $n=71$, and $z=1.96$.

Statistical significance. For pairwise comparison of HR@5 between two systems, McNemar’s test is applied on the per-query hit/miss binary vectors: $\chi^2 = (b - c)^2 / (b + c)$, where b and c count queries on which only the first or only the second system hits. Results appear in Table 1.

6.2 Dimension 2: Reliability

Per-query hit patterns across the three conditions are tabulated in Table 3. We report: (i) queries where all three systems hit; (ii) queries where the hybrid strictly improves over *both* baselines; (iii) queries where the hybrid regresses versus at least one baseline; (iv) queries that all three systems fail (*hard queries*). Hard queries define the ceiling for first-stage retrieval performance under the current corpus and are discussed in §9.

The reliability dimension directly addresses RQ2: a system that achieves high average HR@5 but exhibits a large hard-query tail is insufficient for deployment in legal AI settings where every query must return a citable provision.

6.3 Dimension 3: Efficiency

Table 2 reports index size on disk (MB), index build time (seconds), median query latency (ms), and 95th-percentile query latency (ms) for each condition, together with the overhead ratio of hybrid latency to the faster single-retriever latency.

7 Results

Tables 1–4 report the full experimental results. All metric cells are marked [PLACEHOLDER] pending completion of the evaluation runs; the table structures are final.

Table 1: Retrieval effectiveness ($n=71$ queries). Wilson 95 % confidence intervals in parentheses for HR@ k columns. † McNemar’s p vs. BM25-only; ‡ vs. BGE-M3-only.

System	HR@1	HR@5		
BM25-only	[PLACEHOLDER]	[PLACEHOLDER]	([PLACEHOLDER], [PLACEHOLDER])	
BGE-M3-only	[PLACEHOLDER]	[PLACEHOLDER]	([PLACEHOLDER], [PLACEHOLDER])	
BM25+BGE-M3 RRF	[PLACEHOLDER]	[PLACEHOLDER]	([PLACEHOLDER], [PLACEHOLDER])	

Table 2: Efficiency. Build time and latency measured on [hardware]. Latency: median and P95 over 5 runs $\times$ 71 queries. Overhead ratio: hybrid median $\div$ faster single-retriever median.

System	Index size (MB)	Build time (s)	Median lat. (ms)	P95 lat. (ms)
BM25-only	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
BGE-M3-only	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
BM25+BGE-M3 RRF	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Overhead ratio	—	—	[PLACEHOLDER]	[PLACEHOLDER]

Table 3: Reliability breakdown ($n=71$ queries). “Hybrid strictly improves” = hybrid hits; both baselines miss. “Hybrid regresses” = hybrid misses; at least one baseline hits.

Category	Queries (n)	Fraction
All three systems hit	[PLACEHOLDER]	[PLACEHOLDER]
Hybrid strictly improves over both bases	[PLACEHOLDER]	[PLACEHOLDER]
Hybrid regresses vs. ≥ 1 baseline	[PLACEHOLDER]	[PLACEHOLDER]
All three systems miss (hard queries)	[PLACEHOLDER]	[PLACEHOLDER]

Table 4: Ablation: sensitivity of hybrid RRF to the k constant (Equation 1). HR@5 and MRR@5 on the full 71-query benchmark. Baseline: $k=60$.

k	HR@5	MRR@5
10	[PLACEHOLDER]	[PLACEHOLDER]
30	[PLACEHOLDER]	[PLACEHOLDER]
60	[PLACEHOLDER]	[PLACEHOLDER]
100	[PLACEHOLDER]	[PLACEHOLDER]

8 Discussion

RQ1: Hybrid effectiveness and reliability. Table 1 will report HR@5, HR@10, MRR@5, and MRR@10 for all three retrieval conditions, together with McNemar’s p -values for pairwise comparisons. The central prediction is that BM25+BGE-M3 RRF outperforms both individual baselines on HR@5, consistent with the independence hypothesis: BM25 is expected to fail on vocabulary-mismatch queries that BGE-M3 can recover, while BGE-M3 is expected to scatter across many definitional articles that BM25 anchors via exact-term match. The reliability breakdown in Table 3 will quantify whether this complementarity manifests as a reduced hard-query rate. These are testable predictions that will be confirmed or refuted once the experimental runs complete.

RQ2: Sufficiency for grounded responses. A non-zero hard-query rate (Table 3) would indicate that for some legal questions, retrieval fusion alone cannot surface the relevant provision—either because it is absent from the 72-instrument corpus or because the query requires cross-instrument reasoning that flat retrieval cannot resolve. In a deployed grounded legal AI system, such queries should trigger an insufficient-context guard, preserving attribution reliability at the cost of coverage. Whether the proportion of hard queries is low enough to make the hybrid a viable production first-stage retriever for EU Climate Law is the empirical question RQ2 poses; the answer will be available once the evaluation runs complete.

Comparison with prior work. Prior work on hybrid retrieval in the legal domain [11, 10] has generally motivated fusion-based approaches as a way to reduce failure rates relative to single-retriever baselines, though on different collections and jurisdictions from those studied here. The extent to which these motivations hold for EU statutory text is an open empirical question that our results will help address. The efficiency data in Table 2 will quantify whether concurrent execution holds the latency overhead of hybrid retrieval to an acceptable multiple of single-retriever latency for interactive legal research applications.

Practical implication. At a fusion cutoff of $k_f=5$, first-stage hybrid retrieval must surface at least one correct CELEX instrument within five positions for the downstream generator to produce a citation-supported answer. If coverage is sufficient, the hybrid can serve as a production first-stage retriever; if the hard-query tail is large, corpus expansion or a cross-encoder re-ranking

step is required before deployment. The ablation in Table 4 will indicate whether the standard $k=60$ constant is robust to perturbation for this collection size.

9 Error Analysis

We characterise five anticipated failure modes based on the theoretical properties of BM25 and BGE-M3 and the structural characteristics of the EU Climate Law corpus. Each case is drawn from a representative query in the gold benchmark; the expected CELEX instrument follows from the relevance judgements; the failure mechanism is a theoretical prediction. Actual retrieved results are marked [PLACEHOLDER] and will be populated from the experimental runs. If the observed retrieval pattern for a case does not match the predicted failure mode, the characterisation will be revised accordingly.

Case 1: Vocabulary mismatch (predicted BM25 failure). *Query:* “What monitoring obligations apply to aircraft operators under the EU ETS?” *Expected:* 32003L0087 (Articles 14–15). *BM25 top-1:* [PLACEHOLDER] (predicted: a maritime MRV instrument, due to term-frequency overlap on *monitoring, operators, obligations*). *BGE-M3 and hybrid top-5:* [PLACEHOLDER]. *Predicted failure mode:* vocabulary mismatch. *Implication:* domain-specific query expansion for aviation ETS vocabulary would reduce this class of BM25 failures without requiring model fine-tuning.

Case 2: Semantic drift (predicted BGE-M3 failure). *Query:* “What is the definition of ‘installation’ under the EU ETS?” *Expected:* 32003L0087 (Article 3). *BGE-M3 top-5:* [PLACEHOLDER] (predicted: definitions articles from CBAM, the Energy Efficiency Directive, and LULUCF, all of which define *installation*). *BM25 top-1:* [PLACEHOLDER] (predicted: 32003L0087 via exact-term match on “installation” in Article 3 heading). *Predicted failure mode:* semantic drift across many definitions articles. *Implication:* article-type metadata (definitions vs. obligations) could prioritise definitions-article provisions in re-ranking.

Case 3: Article fragmentation (predicted failure for all systems). *Query:* “What criteria govern the free allocation of allowances to industrial installations at risk of carbon leakage in ETS Phase 4?” *Expected:* 32003L0087 and the Phase 4 benchmarks regulation. *All systems:* [PLACEHOLDER] (predicted: individual articles from one instrument appear in top-5 but the cross-instrument answer spans provisions not simultaneously present within $k_f=5$). *Predicted failure mode:* article fragmentation. *Implication:* multi-hop retrieval across instruments, or an increased k_f , is required to support cross-instrument legal questions.

Case 4: Cross-reference gap (predicted hybrid partial failure). *Query:* “How does the Carbon Border Adjustment Mechanism interface with EU ETS Phase 4 for electricity imports?” *Expected:* 32023R0956 (CBAM Regulation) and 32003L0087 (ETS Directive). *Hybrid top-5:* [PLACEHOLDER] (predicted: CBAM articles dominate; ETS cross-references fall at rank 6+, outside $k_f=5$). *Predicted failure mode:* cross-reference gap. *Implication:* increasing k_f or introducing explicit cross-reference expansion would recover the second relevant instrument.

Case 5: Governance-level definitional query (predicted BM25 failure). *Query:* “What reporting period applies to national greenhouse gas inventories under the European Climate Law?” *Expected:* 32021R1119 (European Climate Law). *BM25 top-1:* [PLACEHOLDER] (predicted: a sectoral MRV regulation scoring high due to repeated co-occurrence of *reporting period* and *greenhouse gas*). *BGE-M3 and hybrid top-5:* [PLACEHOLDER]. *Predicted failure mode:* high-frequency competing terms in sector-specific instruments outscoring the governance instrument.

Implication: governance-level queries favour dense retrieval; BM25 may require instrument-scoped search or query rewriting to resolve ambiguity with sector-specific regulations.

10 Threats to Validity

10.1 Internal Validity

The gold benchmark was constructed by the paper’s authors, creating a risk of selection bias toward queries whose structure aligns with the system’s retrieval strengths. A single annotator produced all 71 relevance judgements; no inter-annotator agreement coefficient is reported. Independent annotation of a 20 % sample is required before the benchmark can be considered suitable for primary comparison in published IR evaluation campaigns.

10.2 External Validity

Results are obtained on a single domain (EU Climate Law) and may not transfer to other bodies of EU legislation (competition law, GDPR, financial regulation) or to other jurisdictions. The corpus covers English-language text only; EU legislation is legally binding in 24 official languages and translation effects on retrieval are unknown. The corpus constitutes a snapshot of the CELLAR repository and does not reflect subsequent amendments, consolidated versions, or new instruments enacted after the extraction date. 72 CELEX identifiers cover the core EU climate *acquis* but exclude numerous delegated and implementing acts.

10.3 Construct Validity

Binary relevance cannot distinguish between highly material and marginally relevant instruments; two instruments judged relevant for the same query may differ substantially in how directly they answer it. CELEX-level matching does not assess whether the retrieved *article* within the instrument is the most relevant. Retrieval quality is a necessary but not sufficient proxy for grounded response quality: a system could surface the correct article but still generate a hallucinated answer.

Regarding context-window truncation: the maximum article in the corpus (137,713 characters, $\approx 34,000$ tokens) far exceeds BGE-M3’s 8,192-token limit. The `sentence-transformers` library silently truncates these inputs. The number of articles affected and the resulting impact on dense retrieval quality for this tail have not been separately quantified.

10.4 Reproducibility

All library versions are pinned in `requirements.txt`. The BGE-M3 model is loaded by name (`BAAI/bge-m3`); the weight hash should be pinned at download time to guard against upstream model updates. All hyperparameters are centralised in `config.py` and overridable via environment variables. The generator (Llama 3.3-70B via Ollama) is used only for qualitative demonstration; all quantitative metrics are retrieval-only and do not depend on the generator. Code and benchmark will be released at [repository URL].

11 Conclusion

RQ1. [PLACEHOLDER] (to be answered from Tables 1 and 3 once experimental runs complete).

RQ2. [PLACEHOLDER] (to be answered from the hard-query rate in Table 3).

The central contribution of this paper is an evaluation framework and benchmark that did not previously exist for EU Climate Law statute-level retrieval, together with a fully reproducible hybrid retrieval pipeline that requires no proprietary API keys. The most important current

limitation is that relevance judgements were produced by a single annotator and have not been independently validated, constraining the generalisability of reported metrics.

Future work.

1. **Graded relevance annotation.** Adding three-level relevance judgements (0/1/2) to a subset of the benchmark would enable more granular gain analysis.
2. **Cross-encoder re-ranking.** Deploying a cross-encoder on the top-5 hybrid output would test directly whether re-ranking closes the gap on hard queries.
3. **Multilingual extension.** Extracting parallel French and German versions of the same 72 instruments from CELLAR would enable the first multilingual EU Climate Law retrieval evaluation, exploiting BGE-M3’s multilingual capability.

12 Generative AI Disclosure

Claude Sonnet (Anthropic) was used to assist with code scaffolding for the retrieval pipeline and with drafting portions of this manuscript. GitHub Copilot assisted with boilerplate Python. Llama 3.3-70B via Ollama was used for qualitative generation examples in the system demonstration. All scientific claims, experimental designs, relevance judgements, and reported results were formulated and verified by the authors. No AI tool contributed to the construction of the gold standard or the interpretation of results.

References

- [1] S. Robertson and H. Zaragoza, The probabilistic relevance framework: BM25 and beyond, *Foundations and Trends in Information Retrieval* **3**(4) (2009), 333–389.
- [2] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian and Z. Liu, BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation, in: *Findings of the Assoc. for Computational Linguistics: EMNLP 2024*, ACL, 2024.
- [3] G.V. Cormack, C.L.A. Clarke and S. Buettcher, Reciprocal rank fusion outperforms Condorcet and individual rank learning methods, in: *Proc. 32nd Annual International ACM SIGIR*, ACM, New York, 2009, pp. 758–759.
- [4] T. Mori and Y. Kano, Statute-level retrieval for legal question answering using hierarchical document representations, in: *Proc. JURIX 2023*, IOS Press, 2023.
- [5] H.-T. Nguyen et al., NOWJ at COLIEE 2023: Multi-task and ensemble approaches in legal information retrieval and entailment, in: *Proc. COLIEE 2023*, IOS Press, 2023.
- [6] J. Rayo, I. González-Dios and X. Saralegi, Dense retrieval for regulatory compliance: challenges in applying neural embeddings to EU legislative text, in: *Proc. NLLP Workshop 2024*, ACL, 2024.
- [7] R. Mahari, J. Sheridan and A. Pentland, LegalRAG: A retrieval-augmented generation framework for jurisdiction-aware legal question answering, in: *Proc. NLLP Workshop 2024*, ACL, 2024.
- [8] T. Formal, B. Piwowarski and S. Clinchant, SPLADE: Sparse lexical and expansion model for first stage ranking, in: *Proc. 44th Annual International ACM SIGIR*, ACM, New York, 2021, pp. 2288–2292.

- [9] T. Formal, C. Lassance, B. Piwowarski and S. Clinchant, From distillation to hard negative sampling: making sparse neural IR models more effective, in: *Proc. 45th Annual International ACM SIGIR*, ACM, New York, 2022, pp. 2353–2359.
- [10] A. Louis et al., HyPA-RAG: A hybrid parameter adaptive retrieval-augmented generation system for AI legal and policy applications, in: *Proc. NLLP Workshop 2024*, ACL, 2024.
- [11] J. Savelka and K.D. Ashley, LRAGE: A framework for evaluating legal retrieval-augmented generation systems, in: *Proc. JURIX 2024*, IOS Press, 2024.
- [12] R.O. Weber and L. Soh, CBR-RAG: Case-based reasoning augmented retrieval-augmented generation for legal question answering, in: *Proc. JURIX 2023*, IOS Press, 2023.
- [13] D. Trautmann, A. Petrova and V. Tresp, LexRAG: Lexical retrieval-augmented generation for legal reasoning tasks, in: *Proc. EMNLP 2024*, ACL, 2024.
- [14] J. Niklaus et al., LexDrafter: Assisting EU legislative drafting using retrieval-augmented generation, in: *Proc. JURIX 2024*, IOS Press, 2024.
- [15] K. Branting et al., Exploiting legal document structure for enhanced retrieval-augmented generation, in: *Proc. JURIX 2024*, IOS Press, 2024.
- [16] A. Ravichander et al., Towards reliable retrieval in legal retrieval-augmented generation, in: *Proc. NLLP Workshop 2024*, ACL, 2024.
- [17] L. Zheng, N. Guha, B.R. Anderson, P. Henderson and D.E. Ho, Reasoning-focused retrieval for legal question answering, in: *Proc. EMNLP 2024*, ACL, 2024.
- [18] N. Reimers and I. Gurevych, Sentence-BERT: Sentence embeddings using siamese BERT-networks, in: *Proc. EMNLP 2019*, ACL, 2019, pp. 3982–3992.